

MOHAMED ALMEFREJ

Software Engineer / Data Platform Engineer

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PROFILE

Software Engineer / Data Platform Engineer with experience building data-intensive systems, integrations, automation and scalable platform solutions. Currently working at Norrköpings kommun with Microsoft Fabric, Python, PySpark, SparkSQL, SQL, Delta Lake and Power BI. Strong focus on robust architecture, reusable code, metadata-driven development, data quality and operational reliability.

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL, HTML, CSS, Git, C#, Java, FastAPI, Typer

Data & Platforms: Microsoft Fabric, PySpark, SparkSQL, Delta Lake, ETL/ELT, Data Modeling, Power BI, Pandas, PyArrow

Integration & Automation: APIs, OAuth2, Fabric Pipelines, Azure DevOps, Docker, Docker Compose, GitHub Actions, Microsoft 365

Ways of Working: System Design, Agile/Scrum, monitoring, error handling, testing, CI, documentation, troubleshooting

PROFESSIONAL EXPERIENCE

Norrköpings kommun - Software Engineer / Data Platform Engineer

Aug 2024 - Present | Norrköping, Sweden

- Build and improve technical solutions for the municipality's data platform, integrations and analytics needs.
- Develop ETL/ELT workflows using Python, PySpark, SparkSQL, SQL and Delta Lake.
- Design metadata-driven logic for ingestion, transformation, orchestration and monitoring of data flows.
- Integrate data from databases, files and APIs and work with bronze/silver/gold architecture in Microsoft Fabric.
- Support Power BI reporting through structured data models, reliable pipelines and quality-assured data layers.

Addima AB - Junior API Developer

Sep 2023 - Dec 2023 | Stockholm, Sweden

- Built API integrations between Zoho and other cloud-based systems.
- Contributed to automated workflows and worked closely with the digitalization team on integration logic and APIs.

SELECTED PROJECTS

Delta Incident Inspector Lab - Built a Dockerized Delta Lake incident investigation tool with both a Typer CLI and Streamlit browser UI for comparing table versions, schemas, row counts and business keys, including duplicate-key detection and Markdown incident reports. Technologies: Python, Docker, Delta Lake, Streamlit, Pandas, PyArrow, Pytest, Ruff, GitHub Actions.

Metadata-driven ETL Framework - Contributed to a metadata-driven framework for ingestion, transformation, persistence, monitoring and error handling in Microsoft Fabric. Technologies: Python, PySpark, SQL, Delta Lake, Fabric Pipelines.

Power BI / Fabric Metadata Catalog - Worked with metadata for workspaces, semantic models, tables, columns, relationships and permissions to improve visibility, governance and dependency understanding.

Live CV Website - Designed and built a responsive bilingual live CV website using HTML, CSS and JavaScript, including project filtering, dark/light mode, downloadable CV and a GitHub/Cloudflare Pages workflow.

FastAPI User Management System - Built a backend user-management API with FastAPI and SQLAlchemy, focusing on CRUD workflows, database modeling and REST API structure.

Weather Data Extraction & Database Management App - Created a Python and SQL-based workflow for extracting, structuring and storing weather data for later analysis.

GIS & Travel Behavior Analysis Projects - Analyzed GPS and geographic data using GIS and web mapping tools, including QGIS and Leaflet, to model movement patterns and create thematic map visualizations.

YouTube API GUI - Built a Tkinter-based Python GUI that retrieves YouTube video and channel data such as views, likes, comments and recent videos through API calls.

EDUCATION

Dalarna University - Bachelor of Science, Main Field of Study: Informatics

2020 - 2024 | Borlänge, Sweden

- Focus on system development, databases, IT systems, project work and digitalization.

LANGUAGES & ADDITIONAL

Swedish: fluent / native-level | **Arabic:** native | **English:** professional working proficiency

Additional experience: Customer service, retail, reception, care and service roles 2016-2023. Driving licence B.